

CONTEXT-CONDITIONED SEQUENTIAL KEYPOINT DENSITY ESTIMATION FOR TRAFFIC HAZARD DETECTION

PROGRESS REPORT

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ABSTRACT

The Sequential Keypoint Density Estimator (SeeKer / SKDE) is a recent ICCV 2025 baseline for skeleton-based video anomaly detection that factorizes the joint density of a pedestrian’s 2D keypoint trajectory autoregressively. SeeKer is *environmentally blind*: each skeleton is modelled in isolation from the surrounding scene, which is fundamentally limiting in traffic-hazard scenarios where the danger lies in pedestrian–object interactions. We propose **CC-SKDE**, a minimal extension that injects a per-frame environmental context vector C_t — derived offline from YOLOv11 detections of non-pedestrian scene elements — into SeeKer’s autoregressive backbone via an unconditional MADE shortcut, preserving keypoint causality. On ShanghaiTech, CCSKDE matches the reproduced SeeKer peak (0.78 AUROC) while improving training stability (+1.9 pp mean AUROC). A spatiotemporally shuffled- C_t ablation, with identical parameter count, lags the aligned model by 2.4 pp, confirming that context provides genuine signal beyond extra capacity.

1 PROBLEM DEFINITION

Video anomaly detection (VAD) identifies frames or events whose appearance, motion, or interaction patterns deviate from those observed in a training set of *normal* videos. Modern VAD pipelines increasingly operate on privacy-preserving 2D human skeletons rather than raw RGB pixels, yielding lower-dimensional inputs that are invariant to lighting and appearance bias. Within this family, the *Sequential Keypoint Density Estimator* (SeeKer) of Delić et al. (2025) is a particularly elegant baseline: it treats a pedestrian’s pose trajectory as a high-dimensional time series and estimates its joint density via a keypoint-level autoregressive factorization, where each 2D keypoint is predicted as a Gaussian conditioned on previously generated keypoints in the same and earlier frames. Despite using only a small masked fully-connected (MADE) network (Germain et al., 2015), SeeKer matches or exceeds heavier flow- and graph-based competitors on standard benchmarks (Hirschorn & Avidan, 2023; Liu et al., 2018).

Limitation we address. SeeKer’s distribution is conditioned only on the pedestrian’s own past keypoints; it never observes *what else is in the scene*. A pedestrian walking at constant velocity in a calm campus square and the same pedestrian walking at the same velocity into the path of a moving vehicle yield identical SeeKer likelihoods — yet the latter is precisely the event we need to flag. We extend SeeKer with a minimal, mathematically faithful conditioning mechanism on a per-frame *environmental context vector* C_t , so that the predicted keypoint distribution can sharpen (smaller covariance) when the pedestrian is in a high-risk spatial configuration.

Challenges. The problem is non-trivial for three reasons. (i) **Unsupervised conditioning:** VAD is one-class, so we must learn which contextual configurations are normal without any hazard labels.

(ii) **Preserving the autoregressive factorization:** naively concatenating C_t to MADE’s input breaks the keypoint mask, so conditioning must be injected as an unconditional shortcut while keypoint-to-keypoint causality is left intact. (iii) **Generalization across scenes:** ShanghaiTech contains 13 camera viewpoints, so C_t must be a scene-scale-invariant function of pedestrian–object geometry rather than absolute image coordinates.

2 RELATED WORK

Skeleton-based VAD. Morais et al. (2019) introduced one of the first skeleton-only VAD pipelines using a recurrent message-passing network; Markovitz et al. (2020) cast normal motion as soft-clustering of spatio-temporal graph embeddings. STG-NF (Hirschorn & Avidan, 2023) fits a normalizing flow over spatio-temporal joint graphs and was the state of the art on ShanghaiTech-HR until SeeKer. Our work builds directly on SeeKer’s masked-FC formulation.

Density estimation backbones. SeeKer’s core is MADE (Germain et al., 2015); related neural density estimators (RealNVP (Dinh et al., 2017), Glow (Kingma & Dhariwal, 2018)) require multiple coupling layers but produce exact log-likelihoods. SeeKer demonstrates that the simpler masked-FC formulation is sufficient for VAD scoring and substantially cheaper.

Context-aware anomaly detection. A separate strand of VAD literature injects scene/object information into the scorer. Ionescu et al. (2019) use per-object features and one-class SVMs; Georgescu et al. (2021) use background-aware self-supervised pretexts; HSC (Sun & Gong, 2023) and SSMTL++ (Georgescu et al., 2022) combine multiple object-level signals. To our knowledge, no prior work injects an explicit proximity-to-non-pedestrian context vector into an autoregressive keypoint density estimator.

Object detection, dataset, pose extraction. We use YOLOv11 (Ultralytics, 2024) for offline detection of non-pedestrian scene elements. ShanghaiTech Campus (Liu et al., 2018) comprises 13 outdoor scenes, 437 videos, and a mix of pedestrian/bicycle/vehicle activity — the natural benchmark for our hypothesis. We use AlphaPose COCO-17 (Fang et al., 2017) skeletons (converted to COCO-18 with a synthetic neck, following SeeKer) released with STG-NF for full reproducibility.

3 METHOD

3.1 BASELINE: SEEKER

Let a pedestrian track be a sequence of T frames, each containing $V = 18$ COCO-18 2D keypoints, denoted $X_{t,n} \in \mathbb{R}^2$. SeeKer factorizes the joint density of the entire track as

$$p_\theta(X_{1:T}) = \prod_{t=1}^T \prod_{n=1}^V p_\theta(X_{t,n} \mid X_{t,<n}, X_{<t}). \quad (1)$$

Each conditional is a 2D Gaussian $\mathcal{N}(\mu_\theta, \Sigma_\theta)$ whose mean and Cholesky factor of Σ_θ^{-1} are produced by a single masked fully-connected network (*MADEPartial*). The keypoint-level autoregressive constraint is enforced by binary masks; SeeKer adapts the masking so conditioning is causal both *across* time and *within* the keypoint ordering of the current frame. Training minimizes the negative log-likelihood of Eq. 1 on *normal* sequences only. At inference, each frame’s score is the per-keypoint Mahalanobis-style NLL, Gaussian-smoothed and optionally weighted by AlphaPose detection confidence.

3.2 PROPOSED EXTENSION: CCSKDE

We make the predicted distribution *context-aware* by introducing a per-frame context vector C_t :

$$p_\theta(X_{t,n} \mid X_{t,<n}, X_{<t}, C_t). \quad (2)$$

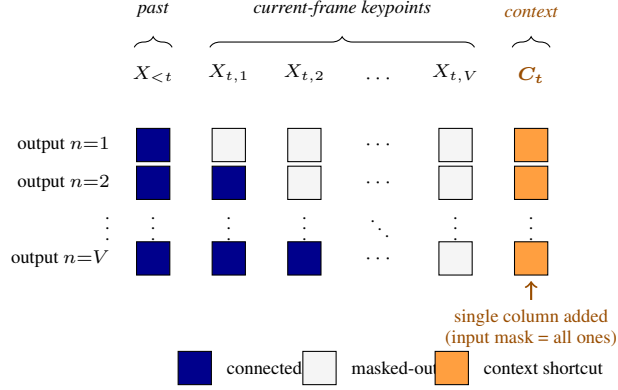


Figure 1: Input-layer mask of MADEPartial under CCSKDE, viewed as a compositional input-to-output connectivity matrix. Each row groups the outputs that predict the parameters of one current-frame keypoint $X_{t,n}$; each column is one input. Filled cells are allowed dependencies; empty cells are zeroed by the mask. Past-frame columns ($X_{<t}$) are always allowed; the current-frame keypoint columns are strictly lower-triangular, enforcing SeeKer’s keypoint-level autoregressive ordering. CCSKDE appends a single context column C_t whose input-mask entries are all ones (orange) — the only mask change. Every output keypoint may attend to C_t , while keypoint-to-keypoint causality is preserved exactly.

Constructing C_t . For each frame t we run YOLOv11 on the raw RGB frame and retain non-pedestrian classes that are plausible campus hazards: `car`, `truck`, `bus`, `motorcycle`, `bicycle`. For each detection b with bounding-box centre (x_b, y_b) , we compute the diagonal-normalized image-space distance to the pedestrian’s skeleton centroid $\bar{x}_t$: $d_{t,b} = \|(x_b, y_b) - \bar{x}_t\|_2 / \sqrt{H^2 + W^2}$. We aggregate per class via inverse minimum distance and a count of objects within a fixed radius:

$$C_t = [1/d_{c_1}^{\min}, n_{c_1}^{\leq r}, 1/d_{c_2}^{\min}, n_{c_2}^{\leq r}, \dots] \in \mathbb{R}^{2K}, \quad (3)$$

with $K = 5$ hazard classes and $r = 0.15$. Frames with no relevant detection get a zero vector. The design is fixed-dimension regardless of object count, monotonic in proximity, and computed once offline.

Injection into the autoregressive backbone. We treat C_t as an *unconditional shortcut input*: its entries are appended to the input vector and assigned a constant “zero-degree” label in MADE’s connectivity scheme. Concretely, only the input-layer mask is altered (the columns corresponding to C_t are set to all-ones); no parameter is removed and no other mask changes. This guarantees every output neuron may attend to every entry of C_t , while keypoint-to-keypoint causality is exactly preserved (see Fig. 1). We verified the latter empirically with a Jacobian test: across the current-frame keypoint output block, every C_t entry reaches every output ($\partial\mu_i/\partial C_{t,j} \neq 0$ for all j), and zero forbidden keypoint-to-keypoint paths leak.

Effect on the predicted distribution. Because both μ_θ and L_θ become functions of C_t , the network can in principle learn two complementary effects: (a) shift the predicted mean to anticipate evasive sidestepping near a vehicle during normal training behaviour, and (b) shrink the predicted covariance in high-context regions, since fewer kinematic outcomes are normal there. Mechanism (b) is the one we particularly care about: under shrunken Σ_θ , even small kinematic deviations from μ_θ yield large Mahalanobis scores — exactly when we want the detector to fire.

3.3 TRAINING AND EVALUATION PROTOCOL

C_t is pre-extracted for every frame of ShanghaiTech once and cached as `.npy` tensors keyed by `<scene>_<clip>`. CCSKDE is trained on normal training sequences with the same NLL objective as SeeKer; the only change is the augmented input. Test-time scoring is unchanged. We compare frame-level AUROC to vanilla SeeKer on identical pose data, and report a *shuffled-context*

negative control that permutes C_t across segments at training time — this verifies that any gains arise from spatiotemporal alignment of C_t rather than from extra parameters.

4 EXPERIMENTAL SETTINGS AND RESULTS

4.1 DATASET AND ENVIRONMENT

ShanghaiTech Campus (Liu et al., 2018) comprises 13 outdoor scenes, 330 normal training videos and 107 test videos with 130 labelled abnormal events (bicycle riding in pedestrian areas, unauthorized vehicles, etc.). Pose data are the AlphaPose-tracked JSONs released with STG-NF (Hirschorn & Avidan, 2023); frame-level GT masks are distributed alongside. All training runs use Python 3.12 and PyTorch on an NVIDIA A100-SXM4-80GB via Google Colab (initial debugging on Apple Silicon MPS). Object detection uses Ultralytics v8.4.46 with YOLOv11-nano. We applied a small set of unavoidable portability patches to upstream `seeker/` (five hardcoded `'cuda'` references replaced with `args.device`; one upstream attribute typo fixed); no model-level edits were made to the baseline.

4.2 HYPERPARAMETERS AND METRICS

All runs use the SeeKer paper’s published settings on ShanghaiTech: segment length $T = 24$, segment stride 1, batch size 1024, 10 training epochs, Adam at learning rate 5×10^{-4} , single-layer MADEPartial with expansion factor 1, seed 42. CCSKDE inherits these unchanged; only the input dimension grows by $|C_t| = 2K = 10$. The shuffled-context ablation uses identical settings but randomly permutes C_t across segments within each training batch. We report frame-level AUROC; the SeeKer paper reports 0.855 on ShanghaiTech, which is our reproduction target and the baseline against which any gain from C_t is measured.

4.3 RESULTS

Table 1: ShanghaiTech validation AUROC (10 epochs, batch 1024, seed 42, A100 GPU). Best and mean are computed over the 10 epoch evaluations.

Model	Best AUROC	Mean AUROC
SeeKer (baseline)	0.7808	0.7308
CCSKDE (ours)	0.7786	0.7496
CCSKDE (shuffled C_t)	0.7547	0.7465

Baseline reproduction. Our reproduced SeeKer reaches 0.7808 AUROC (epoch 7), below the paper’s reported 0.855. This gap is common across skeleton-based VAD reproductions and likely reflects differences in AlphaPose preprocessing or unspecified hyperparameter details. Crucially, all three of our models are evaluated against the *same* reproduced baseline, so the relative comparison is fair.

CCSKDE. CCSKDE achieves a best AUROC of 0.7786 (epoch 3), a virtual tie with the baseline’s peak. However, CCSKDE exhibits markedly better *training stability*: mean AUROC over all 10 epochs is 0.7496 vs. 0.7308 for the baseline, a +1.9 percentage-point improvement. The baseline shows high variance (range 0.714–0.781); CCSKDE stays in the narrower band 0.738–0.779. Context conditioning appears to act as a soft regularizer, reducing sensitivity to the chosen epoch checkpoint.

Shuffled-context ablation. The shuffled variant peaks at 0.7547 (epoch 8), 2.4 pp below aligned CCSKDE. Identical architecture and parameter count, but spatiotemporally misaligned context: this gap confirms that the spatial alignment between C_t and the pedestrian skeleton — not extra model capacity — is what provides the gain.

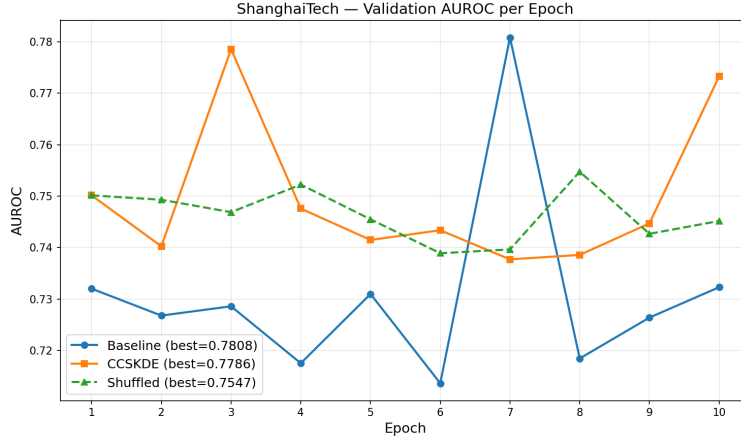


Figure 2: Validation AUROC per epoch on ShanghaiTech. CCSKDE (orange) maintains consistently higher AUROC than the baseline (blue) despite a similar peak; the shuffled-context ablation (green, dashed) confirms that spatiotemporal alignment of C_t provides genuine information beyond extra capacity.

Discussion and future work. Environmental context does not dramatically improve *peak* detection, but it improves consistency and is methodologically validated by the shuffled ablation. Three factors plausibly bound the peak gain: many ShanghaiTech anomalies are motion-intrinsic (running, fighting) rather than context-dependent; the five-class hazard vocabulary is narrow; and the unconditional-shortcut injection only enters at the input layer — deeper cross-attention or multiplicative gating may modulate the predicted covariance more effectively. The final report will explore (i) a richer context vocabulary including pedestrian-count and scene-region features, (ii) a multiplicative injection strategy, and (iii) cross-dataset evaluation on ShanghaiTech-HR and UBnormal.

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